

Azure Cheat Sheet

Baseline: Azure CLI 2.90.0, Bicep 0.47.16, azd 1.34.2 (as of 2026-09-27). Examples use the Bookshelf scenario (\$RG=rg-bookshelf, \$LOCATION=eastus2). Quick reference only — verify against the official docs before production use.

CHOOSING A COMPUTE SERVICE

Service	Best for	Scaling	Scale-to-zero	State	Notes
Container Apps	Microservices, event-driven apps, Dapr	KEDA-based, per-revision	Yes	Stateless	Serverless containers on managed Kubernetes/Envoy
Container Apps Express	Small internal APIs from source, minimal config	Auto	Yes	Stateless	Fastest path via az containerapp up
App Service	Web apps/APIs, standard runtimes	Manual/auto-scale rules	No — plan instances always billed (use Functions Flex/Consumption or Container Apps)	Stateless	Deployment slots, easy custom domains/TLS
Functions	Event-driven, short-lived executions	Auto (Consumption/Flex)	Yes (Consumption/Flex)	Stateless	Triggers/bindings; Flex Consumption for VNet+scale
AKS	Full Kubernetes control, complex topologies	Cluster/node autoscaler, KEDA	With node autoscaler to 0 (user pools)	Either	Automatic SKU reduces ops; use for advanced networking/CRDs
ACI	Single containers, burst/batch jobs	None (per-instance)	N/A (ephemeral)	Stateless	Fastest cold start for one-off container runs
VMs	Legacy/custom OS, licensing constraints	VM Scale Sets	No	Stateful	Full control, most operational overhead

MANAGED DATABASES — ENGINE → AZURE OPTION

Engine	Azure option	Create (essentials)
SQL Server	Azure SQL Database / Managed Instance	az sql server create -g \$RG -n sql-bookshelf -l \$LOCATION --enable-ad-only-auth --external-admin-principal-type User --external-admin-name <upn> --external-admin-sid <object-id> + az sql db create -g \$RG -s sql-bookshelf -n catalog --edition GeneralPurpose --family Gen5 --capacity 2 --compute-model Serverless
PostgreSQL	Azure Database for PostgreSQL flexible server	az postgres flexible-server create -g \$RG -n pg-bookshelf -l \$LOCATION --tier GeneralPurpose --sku-name Standard_D2s_v3 --version 17 --microsoft-entra-auth-enabled --password-auth Disabled
MySQL	Azure Database for MySQL flexible server	az mysql flexible-server create -g \$RG -n mysql-bookshelf -l \$LOCATION --tier GeneralPurpose --sku-name Standard_D2ds_v4
MongoDB	Azure DocumentDB (vCore); Cosmos DB for MongoDB (RUB); MongoDB Atlas (Marketplace)	az cosmosdb mongocluster create -c docdb-bookshelf -g \$RG -l \$LOCATION --administrator-login bookshelfadmin --administrator-login-password <pwd> --server-version 7.0 --shard-node-tier M30 --shard-node-count 1 --shard-node-ha true --shard-node-disk-size-gb 64
NoSQL / Cassandra / Gremlin / Table	Azure Cosmos DB (RU-based APIs)	az cosmosdb create -n docdb-bookshelf -g \$RG --locations regionName=\$LOCATION failoverPriority=0 --default-consistency-level Session
Redis	Azure Managed Redis	az redisenterprise create --cluster-name redis-bookshelf -g \$RG -l \$LOCATION --sku Balanced_B1
Partner / self-managed	Elasticsearch → Elastic Cloud (Azure Native ISV Service, or ECK on AKS; Couchbase → Couchbase Capella (Marketplace) or Autonomous Operator on AKS; Neo4j → Neo4j AuraDB (Marketplace) or Helm on AKS; Qdrant → Qdrant Cloud (Marketplace) or Helm on AKS; Solr / Lucene → self-managed on AKS (Solr Operator)/VMs, or Azure AI Search as the managed alternative. Self-managed engines run on aks-bookshelf with Azure Disk/Files CSI storage.	
Prometheus	Azure Monitor managed service for Prometheus + Azure Managed Grafana	az monitor account create -g \$RG -n amw-bookshelf -l \$LOCATION + az aks update -g \$RG -n aks-bookshelf --enable-azure-monitor-metrics

Entra auth: Azure SQL —enable-ad-only-auth + —external-admin= on create (or az sql server ad-only-auth enable); PostgreSQL/MySQL —microsoft-entra-auth Enabled —password-auth Disabled + —admin-object-id/—admin-display-name/—admin-type; Cosmos DB disable local auth + RBAC data-plane roles; DocumentDB and Managed Redis support Entra ID auth too.

HIERARCHY & REGIONS

Tenant (Entra ID) → **management group** → **subscription** → **resource group** → **resource**. Policies/RBAC assigned at any level, inherited downward.

Region: a set of datacenters with low-latency network.
Availability zone: 1+ physically separate datacenters within a region (min. 3 zones per enabled region). **Region pair:** paired region for DR, usually same geography. Use zone-redundant SKUs (LB, Storage ZRS, zone-redundant AKS node pools) for resilience within a region; paired/secondary regions for region-wide outages.

CLI ESSENTIALS

```
az login
az login --service-principal -u <id> -p <secret> --tenant <tid>
az account list -o table
az account set --subscription <name-or-id>
export RG=rg-bookshelf LOCATION=eastus2
az group create -n $RG -l $LOCATION
az group delete -n $RG --yes --no-wait
```

Output/query: -o table|json|tsv|yaml; —query "[{<name>:name}]<name>}" (JMESPath).

Extensions: az extension add -n containerapp, az extension list, az extension update -n <name>; az upg rade to update the CLI itself.

laC: az deployment group create -g \$RG -f main.bicep -p env=dev; az bicep build; azd up / azd deploy for azd templates.

CONTAINER APPS

```
az containerapp up -n catalog-api --source . \
-g $RG -l $LOCATION --ingress external
az containerapp create -n catalog-api -g $RG \
--environment cae-bookshelf --image \
acrbookshelf.azurecr.io/catalog-api:1.0.0 \
--target-port 8080 --ingress external \
--min-replicas 0 --max-replicas 10
az containerapp update -n catalog-api -g $RG \
--image acrbookshelf.azurecr.io/catalog-api:1.0.1 \
--set-env-vars Kv=V
az containerapp revision list -n catalog-api -g $RG
az containerapp ingress traffic set \
-n catalog-api -g $RG --revision-weight <rev1>=80 <rev2>=20
```

Scale rules: HTTP concurrency, CPU/memory, KEDA scalers (queue length, Kafka lag, cron).

Secrets: az containerapp secret set -n catalog-api -g \$RG --secrets key=value; reference with secretref: key.

Jobs: az containerapp job create —trigger-type Schedule[Event|Manual]. Dapr enabled per app with —enable-dapr —dapr-app-id catalog-api.

ACR & AKS

```
az acr create -n acrbookshelf -g $RG --sku Premium
az acr build -r acrbookshelf \
-t catalog-api:{{.Run.ID}} .
az aks create -n aks-bookshelf -g $RG \
-l $LOCATION --sku automatic
az aks get-credentials -n aks-bookshelf -g $RG
az aks nodepool add -g $RG \
--cluster-name aks-bookshelf \
-n userpool --node-count 3
az aks update -g $RG -n aks-bookshelf \
--attach-acr acrbookshelf
az aks update -g $RG -n aks-bookshelf \
--enable-keda \
--enable-workload-identity --enable-oidc-issuer
az aks upgrade -g $RG -n aks-bookshelf -k <version>
```

Gateway API: enable via —enable-app-routing/AKS-managed NGINX or install the Gateway API CRDs + a supported controller (e.g. Application Gateway for Containers).

Automatic SKU: Microsoft manages node pools, upgrades, security defaults.

BLOB STORAGE

Tiers: Hot > Cool > Cold > Archive (rehydrate needed). **Redundancy:** LRS < ZRS < GRS < GZRS (RA- variants add read access to secondary).

```
az storage account create -n stbookshelf -g $RG \
-l $LOCATION --sku Standard_ZRS --kind StorageV2
az storage blob upload --account-name stbookshelf \
-c product-images -n covers/1.jpg \
-f ./covers/1.jpg --auth-mode login
az storage blob generate-sas \
--account-name stbookshelf -c product-images \
-n covers/1.jpg --auth-mode login \
--as-user --permissions r --expiry <iso>
```

Lifecycle: JSON rules move/delete blobs by age/tier (az storage account management-policy create).

Static website: az storage blob service-properties u pdate —static-website —index-document index.html.

FRONT DOOR & CDN

```
az afd profile create -g $RG \
--profile-name afd-bookshelf --sku Premium.AzureFrontDoor
az afd endpoint create -g $RG \
--profile-name afd-bookshelf --endpoint-name bookshelf
az afd origin-group create -g $RG \
--profile-name afd-bookshelf \
--origin-group-name og-catalog-api \
--probe-request-type GET ...
az afd route create -g $RG --route-name rt-api \
--profile-name afd-bookshelf \
--endpoint-name bookshelf \
--origin-group og-catalog-api \
--https-redirect Enabled
az afd endpoint purge -g $RG \
--profile-name afd-bookshelf \
--endpoint-name bookshelf --content-paths "*"/*"
```

Caching rules: rule set match conditions (path/header/query) → cache-key/cache-duration/compression actions.

KEY VAULT

```
az keyvault create -n kv-bookshelf -g $RG \
-l $LOCATION --enable-rbac-authorization true
az keyvault secret set --vault-name kv-bookshelf \
-n orders-db-connection-string --value "<v>"
az keyvault secret show --vault-name kv-bookshelf \
-n orders-db-connection-string
az role assignment create --role \
"Key Vault Secrets User" --assignee <principal> \
--scope <kv-resource-id>
```

App Service ref: @Microsoft.KeyVault(SecretUri=https://kv-bookshelf.vault.azure.net/secrets/orders-db-connection-string/).

Container Apps: managed identity + secretref backed by a Key Vault secret via —secrets name=keyvaultref:<uri>,identityref:<id>.

CSI driver (AKS): SecretProviderClass mounts secrets as files/synced K8s Secrets.

APP CONFIGURATION

```
az appconfig create -n appcs-bookshelf -g $RG \
-l $LOCATION --sku Standard
az appconfig kv set -n appcs-bookshelf \
--key "catalog-api:retry:max-attempts" \
--value 3 --label dev --yes
az appconfig feature set -n appcs-bookshelf \
--feature CatalogRecommendations --label prod
```

Key Vault reference: store a reference value pointing at a secret URI; App Configuration resolves it via managed identity at read time.

Kubernetes provider: AzureAppConfigurationProvider CRD syncs kv/feature flags into a ConfigMap/Secret for pods.

NETWORKING

```
az network vnet create -n vnet-bookshelf -g $RG \
-l $LOCATION --address-prefixes 10.1.0.0/16 \
--subnet-name snet-apps \
--subnet-prefixes 10.1.0.0/24
az network nsg create -n nsg-apps -g $RG
az network nsg rule create -g $RG \
--nsg-name nsg-apps \
--allow-https --priority 100 \
--destination-port-ranges 443 --access Allow
az network private-endpoint create -n pe-stbookshelf \
-g $RG --vnet-name vnet-bookshelf \
--subnet snet-apps \
--private-connection-resource-id <st-id> \
--group-id blob --connection-name pe-stbookshelf
az network private-dns zone create -g $RG \
-n privateink.blob.core.windows.net
```

GATEWAYS & APIM

Decision: L4 only, cheapest → **Load Balancer** (Standard SKU). L7 HTTP(S), WAF, path routing, single region → **Application Gateway** (v2). Global L7, CDN+WAF → **Front Door**. Full API lifecycle/policies/keys → **APIM**. K8s-native L7 → **Gateway API**.

az apim create -n apim-bookshelf -g \$RG -l \$LOCATION --publisher-name Bookshelf --publisher-email <em ali> --sku-name Developer (v2 tiers via Bicep/az rest).

APIM tiers: Consumption (serverless), Basic v2/Standard v2 (dev/test to prod), Premium (VNet, multi-region, self-hosted gateway).

Key policies: rate-limit-by-key, validate-jwt, set-ba ckend-service, cache-lookup/cache-store, cors.

VPN & HYBRID

```
az network vnet-gateway create -n vgw-bookshelf \
-g $RG --vnet vnet-bookshelf --gateway-type Vpn \
--sku Vpn6w2AZ --vpn-type Routebased --no-wait
az network local-gateway create -g $RG \
--lgw-onprem-bookshelf \
--gateway-ip-address <onprem-ip> \
--local-address-prefixes 192.168.0.0/16
az network vpn-connection create -n conn-onprem \
-g $RG --vnet-gateway1 vgw-bookshelf \
--local-gateway2 lgw-onprem-bookshelf \
--shared-key <key>
# P25 with Entra ID
az network vnet-gateway update -g $RG \
-n vgw-bookshelf --client-protocol OpenVPN \
--address-prefixes 172.16.201.0/24 \
--vpn-auth-type AAD \
--aad-tenant \
https://login.microsoftonline.com/<tid> \
--aad-audience \
632b3d4f-fb67-d084-bdcf-b95ad541b5c8 \
--aad-issuer https://sts.windows.net/<tid>/
az network vnet-gateway vpn-client generate \
-g $RG -n vgw-bookshelf
```

S2S: VNet gateway + local network gateway (on-prem public IP + prefixes) + shared key/IPsec policy.

P2S with Entra ID: OpenVPN tunnel + —vpn-auth-type AAD with tenant/audience/issuer; clients sign in via Azure VPN Client with Entra credentials (MFA/Conditional Access).

EMAIL (ACS)

```
az communication email create -n acs-bookshelf \
-g $RG --location Global \
--data-location "United States"
az communication email domain create -g $RG \
--email-service-name acs-bookshelf \
--domain-name AzureManagedDomain --location Global \
--domain-management AzureManaged
az communication create -n cs-bookshelf -g $RG \
--location Global --data-location "United States"
az communication update -n cs-bookshelf -g $RG \
--set "linkedDomains=[{<domain-id>}]"
```

Link a verified domain (Azure-managed or custom with SPF/DKIM/DMARC) to the Communication Services resource, then send via the **Email SDK/REST API** or the **SMTP relay**: smtp.azurecom.net:587 STARTTLS; username = <acs-resource>.<entra-app-id>.<tenant-id> (e.g. cs-bookshelf.<app-id>.<tenant-id>) or an SMTP Username resource; password = the Entra app's client secret (app needs Communication and Email Service Owner on the resource). Quotas apply per domain/tenant tier (messages/day, recipients/message) — check current limits before high-volume sends.

MESSAGING & INTEGRATION

Service Bus: queues/topics, sessions, dead-lettering, transactions — enterprise messaging.

Event Grid: pub/sub eventing, push delivery, system + custom topics.

Event Hubs: high-throughput event streaming (Kafka-compatible).

Logic Apps: low-code workflow orchestration/connectors.

```
az servicebus namespace create -n sb-bookshelf \
-g $RG -l $LOCATION --sku Standard
az servicebus queue create -n orders-queue \
--namespace-name sb-bookshelf -g $RG
az eventgrid topic create -n egt-bookshelf \
-g $RG -l $LOCATION
```

MONITORING

```
requests
| where timestamp > ago(1h)
| summarize count() by resultCode
exceptions
| where timestamp > ago(24h)
| top 20 by timestamp desc
AppTraces
| where SeverityLevel >= 3
```

OpenTelemetry: instrument with the OTEL SDK/auto-instrumentation and export to Azure Monitor via the azure monitor exporter (or an OTEL Collector) — replaces the legacy Application Insights SDKs.

Alerts on metric/log queries; Action Groups for notification/automation.

GOVERNANCE & COST

```
az policy assignment create -n pol-bookshelf \
--policy <policy-id> --scope <scope>
az lock create -n lock-bookshelf -g $RG \
--lock-type CanNotDelete
az consumption budget create -n budget-bookshelf \
--amount 500 --time-grain Monthly \
--category Cost --start-date <date> --end-date <date>
```

Policy: enforce/audit/deny at any scope; initiatives group related policies. **Locks:** ReadOnly/CanNotDelete, apply above resource level.

Cost: budgets + alerts, Cost Analysis views, reservations/savings plans for committed use discounts.

RETIREMENTS & RENAMES — CHECK DATES BEFORE RELYING ON THEM

- **Basic Load Balancer & Basic Public IP:** retired 2025-09-30 — migrate to Standard SKU.
- **Application Gateway v1:** retirement targeted 2026-04-28 — migrate to v2.
- **Azure Cache for Redis:** being superseded by Azure Managed Redis; legacy tiers on a multi-year retirement path through 2027/2028 — plan migration to Managed Redis.
- **Azure Blueprints:** phased retirement from 2026-07-31, full retirement 2027-01-31 — migrate to Template Specs + Policy/initiatives.
- **Azure Functions Linux Consumption plan:** retirement targeted 2028-09-30 — evaluate Flex Consumption.